

Vibhas Kumar Vats

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Education

2021 – . . .	Ph.D. Computer Science, Indiana University Bloomington, IN.	GPA: 4.0/4.0
2019 – 2021	M.Sc. Data Science with thesis, Indiana University Bloomington, IN.	GPA: 3.97/4.0
2011 – 2015	B.Tech. Electrical Engineering, National Institute of Technology Patna, India.	CGPA: 8.77/10.0

Research Summary

Ph.D. candidate specializing in large-scale machine learning and generative modeling. My research bridges 3D vision, geometry-aware stereo and diffusion, generalizable Gaussian-splatting for novel-view synthesis and scalable neural architectures for representation learning and multimodal synthesis. Experienced in building end-to-end pipelines—from algorithm design to distributed GPU training—and in authoring comprehensive survey work synthesizing the field, with publications at ICCV, WACV, Neurocomputing, SIGSPATIAL, ICCBR, and ACM Computing Surveys.

Selected highlights:

NeurIPS 2026 (under review)	Structured per-pixel Gaussian mixtures for novel-view synthesis (MoG-Splat)
ICCV 2025	Hybrid-Voxel Point-Cloud Upsampling (HVPUNet)
SIGSPATIAL 2025	Diffusion-based geospatial forecasting
WACV 2024	Geometry-consistent multi-view stereo framework (GC-MVSNet)
Neurocomputing 2025	Fusion of 3d geometry and deep learning

Research Publications

Journal Articles

- **V. Vats**, M. A. Reza, D. Crandall, and S.-h. Jung, “Blending 3d geometry and machine learning for multi-view stereopsis,” *Neurocomputing*, p. 131 250, 2025, ISSN: 0925-2312. 🔗 DOI: <https://doi.org/10.1016/j.neucom.2025.131250>.
- **V. K. Vats** and D. J. Crandall, “Geometric constraints in deep learning frameworks: A survey,” *ACM Comput. Surv.*, vol. 57, no. 10, May 2025, ISSN: 0360-0300. 🔗 DOI: 10.1145/3729221.
- C. Wang, M. A. Reza, **V. K. Vats**, et al., “Deep learning-based 3d reconstruction from multiple images: A survey,” *Neurocomputing*, p. 128 018, 2024, ISSN: 0925-2312. 🔗 DOI: 10.1016/j.neucom.2024.128018.

Conference Proceedings

- J. Ha, **V. K. Vats**, M. A. Reza, S.-h. Jung, and D. J. Crandall, “Hvpunet: Hybrid-voxel point-cloud upsampling network,” in *accepted at ICCV*, 2025.
- **V. K. Vats**, D. Varshney, B. Pandey, C. Brelsford, and P. Dias, “Geospatial diffusion for land cover imperviousness change forecasting,” in *accepted at SIGSPATIAL*, 2025.
- **V. K. Vats**, Z. Wilkerson, H. Sato, D. Leake, and D. J. Crandall, “Learning case features with proxy-guided deep neural networks,” in *Case-Based Reasoning Research and Development*, Cham: Springer Nature Switzerland, 2025, pp. 313–327.
- **V. K. Vats**, S. Joshi, D. J. Crandall, M. A. Reza, and S.-h. Jung, “Gc-mvsnet: Multi-view, multi-scale, geometrically-consistent multi-view stereo,” in *Proceedings of the CVF WACV*, Jan. 2024. 🔗 URL: <https://shorturl.at/FUyTf>.
- **V. K. Vats**, Z. Wilkerson, D. Leake, and D. J. Crandall, “Extracting indexing features for cbr from deep neural networks: A transfer learning approach,” in *ICCB*, 2024. 🔗 URL: <https://shorturl.at/qJuwt>.
- Z. Wilkerson, **V. K. Vats**, K. Acharya, D. Leake, and D. Crandall, “Examining the impact of network architecture on extracted feature quality for cbr,” in *CBR Research and Development*, Springer Nature Switzerland, 2023.
- **V. K. Vats** and D. Crandall, “Controlling the quality of distillation in response-based network compression,” in *AAAI International Workshop on Practical Deep Learning in the Wild*, 2022.

Under Review

- **V. K. Vats**, L. Yang, and D. Lungu, *Scale-polytrace: Scalable and efficient building polygonal tracing from satellite imagery*, Under review at WACV, 2027.
- **V. K. Vats**, D. J. Crandall, and S. Goree, *A markovian view of iterative-feedback loops in image generative models: Neural resonance and model collapse*, Under review at Communication AI & Computing, 2026. arXiv: 2602.19033.

- **V. K. Vats**, S.-h. Jung, and D. J. Crandall, *Mog-splat: Structured per-pixel gaussian mixtures for generalizable novel view synthesis*, Under review at NeurIPS, 2026.

Patents and Intellectual Property

- S.-H. Jung, **V. K. Vats**, D. J. Crandall, M. A. Reza, and S. Joshi, "Learning method and device for estimating depth information of image," US Patent App. 18/806,829, Feb. 2025.
- S.-H. Jung, **V. K. Vats**, S. Shubham, M. A. Reza, C. Wang, and D. Crandall, "Method and apparatus for estimating depth information of images," US Patent App. 18/297,396, Oct. 2023.

Technical Skills

Core ML	Diffusion Models, Transformers, CNNs, 3D Reconstruction, Gaussian Splatting, Generative Modeling, Multi-View Stereo, Geometry-aware Deep Learning
Frameworks	PyTorch (advanced), TensorFlow (advanced), CUDA, OpenCV, Keras (advanced)
Languages & tools	Python, R, SQL, Docker, R-Studio, PostgreSQL, Git, HPC, Distributed Training

Work Experience

09/2025-03/2026	Graduate Research Intern , GeoAI group, Oak Ridge National Laboratory (ORNL), Tennessee. - Built and deployed a deep learning pipeline for automated building and road footprint extraction from satellite imagery, achieving $14\times$ faster inference over the prior state-of-the-art - Engineered a polygon connectivity learning module, reducing permutation matching complexity from $O(N^2)$ to $O(K_{max}^2)$ - Reduced memory consumption of pipeline by up to $10\times$, enabling the system to scale to higher-resolution inputs (512×512) and denser scenes previously infeasible in production - Delivered polygon extraction at up to 1200 vertices per image, unlocking richer building geometry detail for downstream GIS and urban planning applications <i>Dataset:</i> Orbital-Net, INRIA, SpaceNet-2, and Massachusetts Roads
Summer-2024	Summer Intern , GeoAI group, Oak Ridge National Laboratory (ORNL), Tennessee. - Collaborated with a multidisciplinary team of researchers to apply AI in geospatial analysis - Developed a generative diffusion model to predict and analyze land use and land cover patterns - Designed a multi-branch conditional normalization technique to control the diffusion model - Implemented large-scale training on distributed GPU clusters ($8\times A100$) for geospatial diffusion <i>Dataset:</i> National Land Cover Database (NLCD)
2017-2019	External Research Fellow , National Institute of Technology Patna, India Project title: Sustainable Smart Grid Framework for Energy Management System Incorporating Available Renewable Resources, funded by the SERB, Government of India - Developed ML model to mitigate the Communication-link failure in load forecasting systems - Developed an electrical load forecasting system using a weighted polynomial regression model
2015-2017	Senior Manager , Tata Motors Ltd. Pantnagar, India. - Optimized maintenance schedule of Generator yard equipment using past maintenance data - Developed SOP for building and maintenance of earthing pits

Research Experience

- 2021-present **Graduate Research Assistant**, Indiana University Bloomington
- ◇ **3D Reconstruction, Generation and Scene Understanding**: Developed an MVS algorithm that enforces multi-view geometric consistency in the end-to-end learning process
 - Exploring the integration of 3D geometric constraints in deep learning-based MVS frameworks
 - Exploring structure-based attention for preserving geometric structure in feature extraction*Datasets*: DTU, Tanks & Temples, BlendedMVS, ETH3D
 - ◇ **Model Collapse in Recursively Trained Generative Models**: Investigating model collapse in recursively trained generative models using a Markov chain framework
 - Developing methodologies to analyze neural resonance phenomena analogous to Alvin Lucier's experiment, within latent space representations*Dataset*: MNIST, ImageNet
 - ◇ **Gaussian Splatting for novel-view generation**: Studying the attributes of diffusion model leading to mode-collapse on recursively trained and generated dataset
Dataset: Objaverse, THuman, Tanks and Temples
- 2020-2025 ◇ **Deep Learning (DL) - Case-Based Reasoning (CBR) Integration**: Developed an algorithm to examine the impact of DL features on CBR models
 - Developed hybrid system leveraging knowledge-engineered and network-learned features together
 - Developed DL algorithms to extract latent representations suitable for CBR via proxy guided functions*Dataset*: AWA2, Flower102, MNIST
- 2020-2021 **Masters thesis** on Response-based Knowledge Distillation ([pdf](#))
 - Analyzed the knowledge distillation process under varying conditions of networks
 - Proposed the soft-label hypothesis to explain the behavior of the distillation process
 - Developed method to pre-train teacher model for effective knowledge distillation

Teaching Experience

- 2022-2025 **Co-instructor/course designer, Computer Vision (CSCI-B657)**, Indiana University Bloomington
- Led deep learning discussions with **Prof. David Crandall** in Spring 2022, 2023, and 2024
 - It extensively covers seminal papers on *CNNs, MLPs, Transformers, Generative Adversarial Networks, Variational Autoencoders*, and *Diffusion Models*.
 - A complete [list of papers](#) covered in Spring'24 and Spring'25

Awards and Achievements

- 2025 **Nominated to Google PhD Fellowship**— one of two students nominated by Indiana University Bloomington
- 2022 **Associate Instructor of the Year**, Indiana University Bloomington
- 2017 **Outstanding Work Award** by Tata Sustainability group for distinctive work in the CSR program.
- 2015 **Best Graduate**, National Institute of Technology Patna, batch of 2015.